

WASP-43b

R-0623

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180209 · created 2026-07-25T12:56:36+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458158.82781 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.012
Transit depth (Rp/Rs)^2	3.92 +/- 0.49 [%]
Orbital Inclination (inc)	79.71 +/- 0.49
Airmass coefficient 1 (a1)	1.121 +/- 0.011
Airmass coefficient 2 (a2)	-0.084 +/- 0.03
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	None
Optimal Aperture	4.75
Optimal Annulus	12.89
Transit Duration (day)	0.0448 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.198 $\pm$ 0.012 vs 0.1594 (deviation 3.22 σ)
Duration fit vs published	0.0448 d vs 0.0512 d (deviation 2.69 σ)
Mid-time O-C	-3.7 min
Depth SNR	16.5
Residual scatter	0.99 %

Light curve

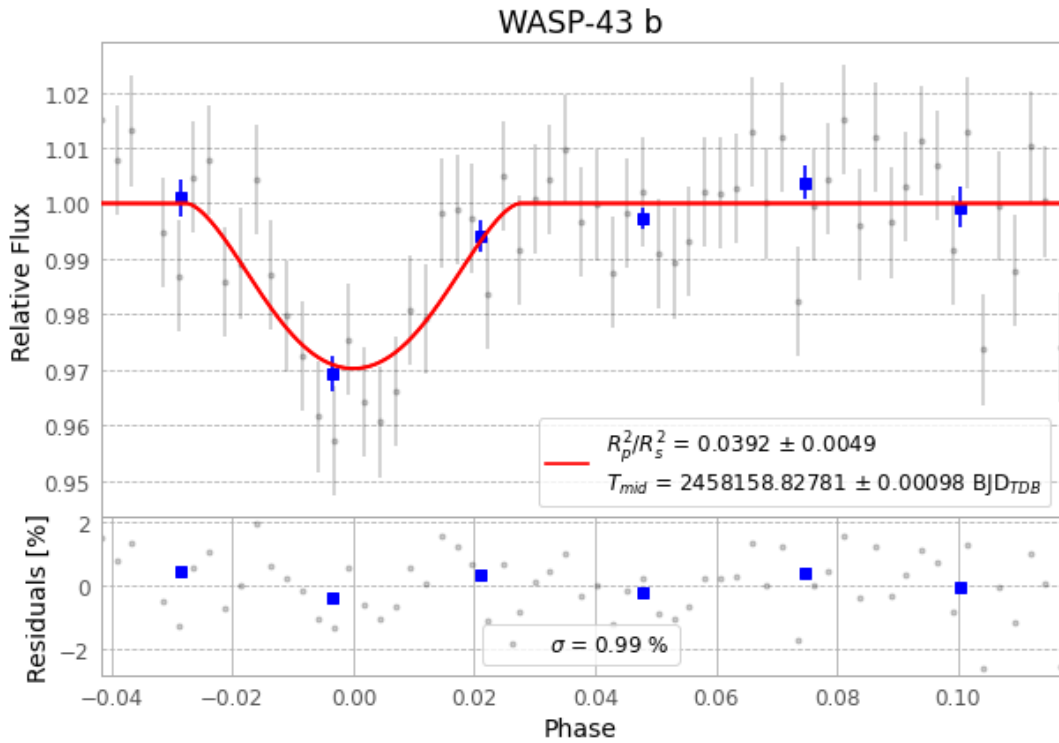


Figure 1 — FinalLightCurve_WASP-43 b_2018-02-08.png (SHA-256 6df82dda72acb27a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [327, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 43a0138f542e2de6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

63 FITS frames (41 MB), WASP-43180209065412.FITS → WASP-43180209100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180209.log (6ba36db41f89...), FinalLightCurve_WASP-43 b_2018-02-08.png (6df82dda72ac...), FinalLightCurve_WASP-43 b_2018-02-08.csv (3657aa84e7fb...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 12:56Z.